

Simple, Flexible, Analytic, Arbitrage Free Option Price Interpolation

Thomas Roos
Quantitative Financial Consulting
thomas.roos@rosabrunetconsulting.co.uk
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Abstract

We introduce a new methodology for interpolating option prices for different strikes at a given expiry. The scheme is asset class independent, arbitrage free, flexible enough to match an arbitrary number of input quotes, and allows for explicit wing control. Computations are fully analytic and interpolated prices are computed at a cost equivalent to evaluating the Black-Scholes formula. Interpolated prices (and implied volatilities) are C^2 smooth. The internal parameters of the model have an intuitive interpretation as a discrete-time local volatility. We also discuss arbitrage free interpolation in expiry.

1 Introduction

The task of interpolating European option prices across strikes at a given expiry is fundamental in derivatives pricing for equities, FX and interest rates. Ideally the interpolation should be arbitrage free by construction and flexible enough to match a large number of input quotes. Interpolated option prices should be fast to calculate and at least C^2 smooth if the construction of a local volatility function is required. Many approaches are described in the literature, the two most popular being the use of parametric implied volatility functions and various model based methods that compute option prices directly.

Parametric implied volatility functions are widely used because they produce fast, analytic option prices and intuitive parameter risk. They typically have a relatively low number of parameters which limits their fitting power, but has the advantage that they produce sensibly shaped density functions. The drawback of these approaches, apart from their limited quote matching ability, is that they are not arbitrage free by construction and that it can be difficult to derive constraints on parameter values that prevent arbitrage. The two most popular approaches in this category are SABR [1] and SVI [2, 3].

The more sophisticated modelling based approaches have the advantage that they are arbitrage free by construction. This category includes exact option pricing formulas like the Heston model [4, 5] as well as various phenomenological models [6, 7, 8]. These approaches can be parametric or include a non-parametric component for extra flexibility. Their disadvantage is that they typically require a numerical step to compute the option

price, for example a one-dimensional numerical integration. This impacts performance and tractability compared with analytic schemes.

A third category of interpolation, which appears to be less popular in practice, constructs convex, non-parametric interpolators in price space [9, 10, 11, 12]. These methods, being non-parametric, have the advantage that, at least in principle, arbitrarily many input prices can be reproduced exactly.¹ One drawback of these schemes is that, without some form of regularisation, there is little control over the shape of the resulting probability density, which can be highly oscillatory due to overfitting. Another drawback is that the internal representation of the model (e.g. weights, spline coefficients, or other numerical parameters) typically does not lend itself to an intuitive interpretation. In addition, it can be unclear how to achieve an arbitrage free interpolation in time to expiry and how to compute sensible risk metrics.

In this paper we introduce a new non-parametric interpolation approach that combines the most desirable features of this category while also addressing some of the shortcomings. The approach is arbitrage free by construction, fully analytic,² flexible enough in principle to match an arbitrary number of input quotes, and C^2 smooth. There is also a simple mechanism for controlling the shape of the resulting distribution, including enforcing uni-modality, and an intuitive way to visualize and interpret the internal representation of the model. The approach amounts to specifying the function that maps the distribution of the underlying to a normal distribution at a discrete set of points, together with a monotonic interpolation scheme and simple, one parameter extrapolation in the wings.

The paper is organised as follows: in Section 2 we introduce the mapping for underlyings with support on the whole real line, and obtain the corresponding option prices in Section 3. Section 4 extends the analysis to asset-like underlyings with support on the positive real line. Section 5 discusses calibration of the models to market option quotes and Section 6 describes how to calculate risk and hedges in this setup. Section 7 derives arbitrage free interpolation in the expiry dimension while Section 8 shows examples of numerical results. We conclude in Section 9.

In essence, the results of this paper are an extension of the results of [11], which arrives at the same interpolation scheme via a different route and which we were unaware of when undertaking this research. A more detailed comparison of the two papers can be found in the conclusion.

2 Rate-like Underlyings

To motivate our approach, recall that, for underlyings such as interest rates that do not have a natural lower bound, the simplest model is to assume normal dynamics of the form

$$dS(t) = u(t)dt + \sigma_N dW(t) . \quad (1)$$

These dynamics produce a marginal distribution that can be written

$$S(T) = F(T) + \sigma_N x \quad (2)$$

¹See [13] for a different non-parametric approach.

²By this we mean that, given the internal parameters of the model, option pricing is fully analytic. All of the schemes mentioned above require a numerical calibration, typically a non-linear least squares optimisation, to calibrate the internal parameters of the model to market quotes.

where $F(T) = E[S(T)]$ is the forward and $x = z\sqrt{T}$, with z being a standard normal variable. We will generalise this formula by letting the volatility term depend on x , that is we will write:

$$S(T) = F(T) + \eta + \sigma_N^{DLV}(x)x, \quad (3)$$

where η is a normalisation constant that ensures $E[S(T)] = F(T)$. Conceptually $\sigma_N^{DLV}(x)$ is a state dependent (or “local”) volatility at expiry T . We will refer to it as the Discrete-time Local Volatility (DLV).³

Not every possible DLV produces a well defined model, and the required consistency condition is most easily expressed in terms of

$$g(x) = \sigma_N^{DLV}(x)x. \quad (4)$$

Then defining $s = S(T) - F(T)$, we have

$$s = \eta + g(x), \quad (5)$$

$$\eta = -E[g(x)] = -\int_{-\infty}^{\infty} dz g(z\sqrt{T})n(z), \quad (6)$$

and the PDF of s is given by

$$\phi(s) = \frac{n\left(\frac{x}{\sqrt{T}}\right)}{g'(x)\sqrt{T}}, \quad (7)$$

where $x(s) = g^{-1}(s - \eta)$ and $'$ denotes the derivative with respect to x . For the model to be well defined, we need $g(x)$ to be monotonically increasing. Then the model is free from butterfly arbitrage by construction since the PDF is positive. Additionally, continuity (C^0) of $g'(x)$ implies continuity of $\phi(s)$, which means the CDF is C^1 and option prices and implied volatilities are C^2 . Since $g(x)$ maps the distribution of the underlying to a normal distribution, we will refer to this construction as the Normal Mapping (NM) approach.

The NM approach then consists of specifying a set of values $\{g_i\}_{i=0}^n$ on a set of knot points $\{x_i\}_{i=0}^n$, both of which are strictly increasing, together with a monotone C^1 interpolation scheme to ensure that $g(x)$ is increasing everywhere and that $g'(x)$ is continuous. In practice we choose monotone cubic interpolation, i.e. $g(x)$ is a cubic polynomial over each interval (see Appendix A for details). Without loss of generality,⁴ and to satisfy (4), we set $g(0) = 0$, so that $s = \eta$ is the median (i.e. $\Phi(\eta) = N(0) = \frac{1}{2}$, where $\Phi(s)$ is the CDF of s).

What remains is to specify the behaviour in the wings, i.e. for $x \leq x_0$ and $x \geq x_n$. The simplest form that ensures continuity of $g(x)$ and $g'(x)$ and allows for additional wing control is

$$g(x) = g_0 + g'(x_0)(x - x_0) + m_l(x - x_0)^2 \quad x \leq x_0 \quad (8a)$$

$$= g_n + g'(x_n)(x - x_n) + m_r(x - x_n)^2 \quad x \geq x_n \quad (8b)$$

The parameters m_l and m_r directly control the behaviour of the implied volatility in the wings, as shown in Appendix B.

³The specification (3) can be viewed as the local volatility version of the Discrete-time Stochastic Volatility model of [7].

⁴A different choice amounts to a redefinition of η .

With this prescription, any increasing sequence of points g_i defines an arbitrage free option pricing model with continuous PDF. We will discuss calibration below, but point out here that, at least in principle, the approach can match *any* non-arbitragible set of input prices. This is because any (continuous) distribution can be mapped to a normal variable via some $g(x)$, and any $g(x)$ can be arbitrarily closely approximated with a dense enough set of knot points.

A given calibrated mapping $g(x)$ can then be visualized by computing the corresponding DLV via (4). The relationship between DLV and implied volatility is conceptually similar to the relationship between the Dupire local volatility and implied volatility. For example, the DLV will exaggerate any wiggles in the implied volatility and large oscillations are an indication that the model is straining to fit the market data. For further insight into the relationship between DLV and implied volatility, see Appendices C and D.

3 Option Prices

The put option price $P(K, T)$ can be written

$$P(K, T) = \int_{-\infty}^{x_K} \frac{dx}{\sqrt{2\pi T}} n\left(\frac{x}{\sqrt{T}}\right) (k - \eta - g(x)) \quad . \quad (9)$$

where $k = K - F$ and $x_K = g^{-1}(k - \eta)$. To compute x_K , we first find $i^* = \max\{i \mid g_i < g(x_K)\}$. Then since $g(x)$ is an increasing cubic over each interval, there is exactly one solution to the cubic equation $g(x) = k - \eta$ for $x \in [x_{i^*}, x_{i^*+1}]$. The root is available analytically (see e.g. [14] on how to compute it robustly).

To calculate the option price, we note that the first two terms in (9) are just $(k - \eta)N(x_K/\sqrt{T})$. Since $g(x)$ is a piecewise cubic polynomial (quadratic in the wings), $g(x) = a_j x^3 + b_j x^2 + c_j x + d_j$ for $x \in [x_{j-1}, x_j]$, the last term is a sum of terms of the form $I(j, x_j) - I(j, x_{j-1})$ where

$$I(j, x_u) = \int_{-\infty}^{x_u} \frac{dx}{\sqrt{2\pi T}} n\left(\frac{x}{\sqrt{T}}\right) (a_j x^3 + b_j x^2 + c_j x + d_j) \quad (10a)$$

$$= (b_j T + d_j)N(z_u) - \left(a_j T^{3/2}(z_u^2 + 2) + b_j T z_u + c_j T^{1/2}\right) n(z_u) \quad , \quad (10b)$$

and $z_u = x_u/\sqrt{T}$. The full option price can therefore be written as

$$P(K, T) = (k - \eta)N\left(\frac{x_K}{\sqrt{T}}\right) - \sum_{j=0}^{i^*+1} (I(j, \min(x_K, x_j)) - I(j, x_{j-1})) \quad . \quad (11)$$

where we have defined $x_{-1} = -\infty$ and $x_{n+1} = \infty$. This analytic formula for the integral is extremely fast to evaluate.⁵

Call prices can be obtained by put-call parity or by writing down the expression equivalent to (9) for the call.

⁵For maximum efficiency, we note that η requires all of the interval integrals, which can then be cached. Computing an option price then only requires a single evaluation of (10) with upper limit z_K , at the same cost as a Black-Scholes option price evaluation.

We note that in addition to option prices, many other useful expectations of functions of s are available in closed form in the NM approach, including all moments. More generally, if a set of functions $f_i(S)$ can be sensibly approximated by polynomial splines⁶, then any payoff of the form $E[\prod_i f_i(S)\mathbb{1}_{S < K}]$ is available in closed form since the product is still a polynomial in z and the indefinite integral is known. This is useful, e.g. when pricing Constant Maturity Swap (CMS) products, where the conditional expectation of (one over) the annuity $E[(A(T))^{-1}|S]$ can be approximated by a spline to give closed form formulas for the values of CMS swaps and options.

4 The Asset Model

For an asset-like underlying with range $(0, \infty)$, the natural approach would be to exponentiate (5). However, with a cubic $g(x)$ this would require a numerical integral to compute option prices, since the integral of the exponential of a cubic is not known in closed form. To retain analytic tractability, we therefore use the fact that the cubic interpolation can approximate the exponential of a function as well as the function itself, and only exponentiate the mapping function (8) in the wings. Specifically, we write

$$s = \zeta h(x) , \quad (12)$$

where $s = S(T)/F$, $\zeta = (\int_{-\infty}^{\infty} dz n(z) h(z\sqrt{T}))^{-1}$ ensures $E[s] = 1$, and we define $h(0) = 1$ so that $s = \zeta$ is the median. Conceptually, we can think of $h(x) = e^{g(x)}$, but we represent $h(x)$ directly in terms of the monotone cubic spline, except in the wings where we write

$$h(x) = e^{a_l + b_l(x-x_0) + m_l(x-x_0)^2} \quad x \leq x_0 \quad (13a)$$

$$= e^{a_r + b_r(x-x_n) + m_r(x-x_n)^2} \quad x \geq x_n . \quad (13b)$$

The constants $a_{l,r}$ and $b_{l,r}$ are determined by making $h(x)$ and $h'(x)$ continuous (i.e. matching the monotone cubic spline at the end points), while m_l and m_r control the asymptotic implied volatilities in the wings as shown in Appendix E.

As before the model is then defined by monotone cubic interpolation on an increasing sequence $\{h_i\}_{i=0}^n$ on a set of knot points $\{x_i\}_{i=0}^n$, except that now $h_i > 0$ and $h(0) = 1$. For visualisation and interpretation purposes, analogous to the rates case, here we define the Lognormal Discrete-time Local Volatility $\sigma_L^{DLV}(s) = \ln(h(x(s)))/x(s)$, which is intuitively a lognormal volatility level at $s(x)$. If $\sigma_L^{DLV}(s) = \sigma_0$ is constant, then s is lognormal with volatility σ_0 and $\zeta = e^{-\frac{1}{2}\sigma_0^2 T}$. We also see from (13) that m_l and m_r are the slopes of σ_L^{DLV} in the wings.

The probability density function (PDF) of s is then given by

$$\phi(s) = \frac{n\left(\frac{x}{\sqrt{T}}\right)}{\zeta h'(x)\sqrt{T}} . \quad (14)$$

The scaled put option price $p(K, T) = P(K)/F$ can be written

$$p(K, T) = \int_{-\infty}^{x_K} \frac{dx}{\sqrt{2\pi T}} n\left(\frac{x}{\sqrt{T}}\right) (k - \zeta h(x)) , \quad (15)$$

⁶And including tails of the form $e^{-\alpha|S|}$.

where $k = K/F$ and $x_K = h^{-1}(k/\zeta)$ can be found by solving the corresponding cubic as before. The integrals required by (15) are the same as before (see (10)), except in the wings where we need

$$I = \frac{1}{\sqrt{2\pi}} \int dz e^{-az^2+bz+c} \quad (16a)$$

$$= \frac{1}{\sqrt{2a}} e^{c+\frac{b^2}{4a}} N\left(\sqrt{2a}z - b/\sqrt{2a}\right), \quad a > 0. \quad (16b)$$

As before, the continuity of $h'(x)$ ensures that the PDF is continuous (C^0), so the CDF is C^1 and option prices and implied volatilities are C^2 . This property is important when constructing a local volatility surface, since the Dupire formula requires the second derivative of option prices with respect to strike.

Finally we note that if the underlying has a lower bound different from zero, one can simply use a shifted version of the asset model, i.e. $s = \zeta h(x) + \delta$, with δ a positive or negative constant as required.

5 Calibration

We begin by discussing the calibration for rate-like (unbounded) underlyings. Starting from an initial guess $\tilde{g}(x)$, the at-the-money (ATM) option price is given by

$$\tilde{P}(K = F, T) = - \int_{-\infty}^{x_F} \frac{dx}{\sqrt{2\pi T}} n\left(\frac{x}{\sqrt{T}}\right) (\tilde{\eta} + \tilde{g}(x)), \quad (17)$$

where $\tilde{g}(x_F) = -\tilde{\eta} = E[\tilde{g}]$. We note that x_F is invariant under a multiplicative scaling $\tilde{g}(x) \rightarrow \beta \tilde{g}(x)$, so $\tilde{P}(F, T)$ is proportional to β . This means that the ATM option price can be calibrated exactly for any given $\tilde{g}(x)$ by simply defining

$$g(x) = \frac{P^{ATM}}{\tilde{P}(F, T)} \tilde{g}(x), \quad (18)$$

where P^{ATM} is the market ATM option price.

Calibration to OTM option prices then proceeds via standard tools, for example the Levenberg-Marquardt or hybrid non-linear least squares algorithm. One subtlety is that instead of calibrating the $\{g_i\}$ for a fixed set of $\{x_i\}$, one wants to calibrate the $\{x_i\}$ for fixed $\{g_i\}$. The reason is that as seen from (5), $\Delta S_i = S_{i+1} - S_i = \Delta g_i$, so the $\{g_i\}$ fix the distance between knots in S-space. In practice one can simply set $g_i = k_i$, which (together with $g(0) = 0$) will place knots approximately at the strikes $K_i = k_i + F$, with the knot spacing fixed in underlying space.⁷ The optimiser then varies the $\{x_i\}$ to minimise the calibration error. This is equivalent to calibrating $x(s)$ instead of $s(x)$. To ensure that the optimiser maintains the $\{x_i\}$ in increasing order, we use $\Delta x_i = x_{i+1} - x_i$ as the free variables and impose the constraint $\Delta x_i > 0$.⁸ Jacobian derivatives $\partial V / \partial x_i$

⁷The scaling (18) changes the knot spacing by the ratio of the put prices β . This is generally not an issue since $\beta \approx 1$ for any reasonable guess $\tilde{g}$, but if there is reason to maintain the spacing exactly, one can always keep $\Delta \tilde{g}$ fixed and set $x_i = \tilde{g}^{-1}(\tilde{x}_i/\beta)$ instead. This will incur a small (typically insignificant) resampling error from the change of interpolation.

⁸The $\{x_i\}$ can be reconstructed from the $\{\Delta x_i\}$ and the condition $g(0) = 0$. In practice it can be useful to impose a constraint $\Delta x_i > \epsilon$, $\epsilon > 0$ to avoid numerical problems when redundant knots are pushed together by the optimiser.

can be computed analytically or via finite differences, but in a modern setup would be available “for free” via Automatic Differentiation (AD).

We now discuss the initial guess $\tilde{g}$. In practice one would typically start the calibration from the previous solution, i.e. set $\tilde{g}$ to the output of the previous calibration. For “cold” starting a calibration, we show in Appendix D that in the short-time limit $\eta = 0$ and the DLV is simply related to the implied volatility, i.e. $\sigma_N^{DLV}(x_K) = \sigma_N(K)$, where $g(x) \equiv x\sigma_N^{DLV}(x)$. Therefore using $\tilde{g}_i = k_i$, $x_i = k_i/\sigma_N(K_i)$ together with (18) is a good initial guess that typically ensures rapid convergence. This approximation amounts to setting the DLV equal to the implied volatilities σ_N and then fixing the overall level by calibrating the ATM.

The calibration for the asset case proceeds along similar lines. Knot ratios are given by $s_{i+1}/s_i = h_{i+1}/h_i$, which together with $h(0) = 1$ suggests $h_i = K_i/F$ for knots placed approximately at K_i . The main difference from the rates case is that there is no exact analytic ATM calibration analogous to (18).⁹ However, we can still approximately calibrate the overall volatility level for a guess $\tilde{h}(x)$ by adding a DLV volatility adjustment equal to the difference $\delta\sigma_{BS}$ between the market and model implied ATM volatilities: $h(x) = \tilde{h}(x)e^{x\delta\sigma_{BS}}$. A simple calibration setup would then set $\tilde{h}_i = k_i$, $x_i = \ln k_i/\sigma_{BS}(K_i)$ as an initial guess, where $k=K/F$ and $\sigma_{BS}(K)$ is the Black-Scholes implied volatility.

We conclude with some general comments about the calibration.

- An important feature of the NM approach is that the mapping function is completely flexible and can match any number of non-arbitragible option quotes.¹⁰ This flexibility can also be a drawback, however, in that there are no in-built constraints on the shape of the resulting PDF $\phi(s)$. If the input data is somewhat noisy, as is often the case e.g. due to asynchronous prices or temporary distortions, matching the option quotes exactly may lead to complex, multi-modal PDFs. While such PDFs are not arbitragible per se, they are often considered unnatural financially and therefore undesirable.¹¹ One may therefore wish to sacrifice some calibration accuracy to maintain a smoother, unimodal PDF. The most direct way to achieve this is to leverage the fact that we can explicitly calculate the location and nature of the extrema of the PDF (see Appendix F). One can then add a penalty for each minimum, for example proportional to the height difference from its smallest neighbouring maximum.¹² By changing the size of this penalty, one trades off calibration accuracy versus unimodality. We show an example of this in the Numerical Results section.
- Even when the target option prices are arbitragible (e.g. due to stale data), the NM calibration will always produce a non-arbitragible model by construction. It therefore provides a simple way to incorporate such quotes into an arbitrage free volatility surface, for example by assigning lower weights to older prices.

⁹More precisely, the mapping to exactly calibrate the ATM has an additive component that shifts the support of the distribution from $(0, \infty)$ to $(1 - \beta, \infty)$, which is generally undesirable for an asset underlying even if $\beta \approx 1$.

¹⁰The mapping can be made as flexible as desired by inserting more knots $\{x_i\}$. There is no restriction that each knot needs to correspond to a calibration strike.

¹¹There are however situations (e.g. short dated expiries around earnings announcements) where multi-modal distributions are financially meaningful, and the NM approach accommodates these cases naturally.

¹²This ensures that a shallow minimum between the main peak and a secondary peak only gets penalised by its depth wrt the latter.

- The wing parameters m_l and m_r can be optimised as part of the calibration, specified exogenously to achieve a certain implied volatility asymptotic (see Appendices B and E), or determined by other criteria (for example, making the second derivative of the mapping function continuous at z_0 and z_n). In the asset case, if there are sufficiently out-of-the-money options available to estimate the asymptotic slopes of the implied volatility, they can also be computed directly from (49).
- An alternative way to initialise the calibration is to use an analytic parametrised model. If the model's CDF is $\Phi_m(S)$, we choose a set of knot points $\{K_i\}$ and compute the corresponding NM via $\eta = \Phi_m^{-1}(0.5) - F$, $g_i = K_i - F - \eta$, $x_i = N^{-1}(\Phi_m(g_i + F + \eta))$. This fixes the initial shape of the PDF and provides a good starting point for the subsequent calibration. We note that this approach can be used even if the analytic model is approximate with possible arbitrage in the wings.¹³ As long as the CDF is increasing over the set of knot points, the NM will produce an arbitrage free model that matches the shape of the original model PDF in the central region. The mapping can then be viewed as an arbitrage free version of the original model, with additional wing control.

6 Risk

Once the $\{x_i\}_{i=0}^n$ are determined by calibration to the market, the value of any trade or portfolio of trades will be a function $V(\{x_i\})$. Choosing a set of hedge instruments $\{C_j\}_{j=0}^m$, we form a portfolio

$$\Pi = V + \sum_{j=0}^m \alpha_j C_j . \quad (19)$$

Then

$$\partial \Pi / \partial \sigma_i = \partial V / \partial \sigma_i + \sum_{j=0}^m \alpha_j \partial C_j / \partial \sigma_i . \quad (20)$$

We would like to determine the hedge ratios α_j so that $\partial \Pi / \partial \sigma_i = 0$. Viewing this as a matrix equation, we can write the solution as

$$\vec{\alpha} = -C^+(\partial V / \partial \vec{\sigma}) , \quad (21)$$

where C^+ is the Moore-Penrose pseudo-inverse of C . Depending on the relative size of n and m , α is then either the solution that minimises the sum of the squared residuals ($n > m$), the solution with the minimum L2 norm ($n < m$ or $n = m$ and C singular), or the unique solution ($n = m$, C invertible).

This approach is easy to implement, but it is somewhat opaque and can lead to risk instabilities in the case when C is not invertible. In that case a more direct way to compute risk is to decouple the risk calculation from the knot points, as follows: let $\{\tilde{C}_j\}_{j=0}^m$ be a set of options with strikes $\{\tilde{K}_j\}$, with corresponding values $\tilde{x}_j$. To compute risk a point j , define a linear perturbation $\delta g(x)$ as the difference between the lines interpolating $g(\tilde{x}_{j-1})$, $g(\tilde{x}_j)$ and $g(\tilde{x}_{j+1})$, with $g(\tilde{x}_j)$ equal to its original value or perturbed to $g(\tilde{x}_j) + \delta \tilde{g}_j$. The resulting $g(x)$ is still cubic (and monotone for a small

¹³The SABR model [1] is a famous example.

enough perturbation), so all formulas still hold and repricing the hedge instruments then produces the risks $\partial \tilde{C}_i / \partial \tilde{g}_j$. This square matrix is invertible (due to the placement of knots at $\{\tilde{x}_j\}$), so $\partial \Pi / \partial \tilde{\sigma} = 0$ is given by $\tilde{\alpha} = -\tilde{C}^{-1}(\partial V / \partial \tilde{g})$. We note that, once the function $g(x; \{\delta \tilde{g}_i\})$ has been implemented, the derivatives with respect to $\tilde{g}$ are available “for free” in an AD setup. The risk to the slopes m_l and m_r can either be computed and managed separately (e.g. as extrapolation risk beyond the range of tradable strikes), or one can determine these parameters as functions of the g_i (e.g. by enforcing continuity of the second derivative).

Finally we note that if one has a favourite parametrised model and would like to generate risk to those parameters, while using NM for the interpolation, one can use a sampled version of the option price changes induced by a parameter change to compute the corresponding risk in NM.

7 Expiry Interpolation

Liquid market quotes are typically available for a discrete set of option expiries. Calibration of the NM at a given expiry T_j will produce a “slice” of option prices $C(T_j, K)$ that is free from Butterfly arbitrage by construction. The question is then how to interpolate prices to some intermediate expiry T , $T_j \leq T \leq T_{j+1}$.

We begin with the case of tradable assets, where there is the additional constraint that the interpolation should be free from calendar spread arbitrage. For proportional dividends, absence of calendar spread arbitrage requires

$$\frac{C(T_j, K_j)}{K_j} \geq \frac{C(T_i, K_i)}{K_i} \quad \text{for } T_j > T_i \quad (22)$$

where $C(T, K)$ is the call price with expiry T and strike K , and $K_j/K_i = F_j/F_i$ (constant moneyness). We will assume that market quotes are such that (22) holds for all calibrated NM slices. Then as shown in [3], for monotonic $\alpha_T \in [0, 1]$, the interpolation

$$\frac{C(T, K)}{K} = \alpha_T \frac{C(T_j, K_j)}{K_j} + (1 - \alpha_T) \frac{C(T_{j+1}, K_{j+1})}{K_{j+1}} \quad (23)$$

with $K_i = (F_i/F_T)K$ is free from both butterfly and calendar spread arbitrage. Moreover, choosing $\alpha_T = (\sqrt{\theta_{j+1}} - \sqrt{\theta_T})/(\sqrt{\theta_{j+1}} - \sqrt{\theta_j})$, where θ_T is any monotonic interpolation of the ATM implied total variance, will ensure that the ATM option price according to (23) is almost exactly consistent with the chosen variance interpolation θ_T .

For non-tradable underlyings like rates, there is no constraint due to calendar spread arbitrage. In this case we want the expiry interpolation to simply produce sensible distributions in a transparent manner. For parametrised models like SABR, this is typically achieved by interpolating the model parameters themselves. For a non-parametric NM, we require a different approach. We propose to define the mapping $g(T, z)$ at the intermediate time T by interpolating the DLVs while imposing a desired interpolation on the ATM volatility. To this end we define

$$\hat{g}(T, x) = \omega_T g(T_j, x) + (1 - \omega_T) g(T_{j+1}, x) \quad (24)$$

and $\hat{\eta} = E[\hat{g}(T, x)]$, and let

$$g(T, x) = \beta(T) \hat{g}(T, x) \quad (25)$$

where (see (18))

$$\beta(T) = \frac{\omega_T P_j^{ATM} \sqrt{\frac{T}{T_j}} + (1 - \omega_T) P_{j+1}^{ATM} \sqrt{\frac{T}{T_{j+1}}}}{\hat{P}^{ATM}(T)}, \quad (26)$$

with

$$\hat{P}^{ATM}(T) = - \int_{-\infty}^{x_F^T} \frac{dx}{\sqrt{2\pi T}} n\left(\frac{x}{\sqrt{T}}\right) (\hat{\eta} + \hat{g}(T, x)), \quad (27)$$

x_F^T is given by $\hat{\eta} + \hat{g}(T, x_F^T) = 0$,¹⁴ and P_i^{ATM} is the market ATM option price at T_i . While the scaling by β is not necessary to keep the interpolation arbitrage free, this construction has the advantage that the ATM normal volatilities inherit exactly the interpolation ω_T , in other words $\sigma_T^{ATM} = \omega_T \sigma_j^{ATM} + (1 - \omega_T) \sigma_{j+1}^{ATM}$. This makes it straight forward to impose the desired ATM volatility interpolation, e.g. a simple linear interpolation or a more sophisticated event model.

8 Numerical Results

Figure 1 shows the results of fitting USD 1Y into 10Y swaptions from November 2016, at a time when there was significant probability of negative rates. The setup is the most basic possible, with one knot per calibration target and m_l and m_r fixed by making the second derivative of $g(x)$ continuous at the end points. Since η is small, the knots coincide closely with the target strikes. All targets are repriced essentially to machine precision. The figure also shows the discrete-time local volatility σ_N^{DLV} and the PDF (on the right-hand axis). We see that in order to exactly match the tightly spaced calibration targets, the PDF has to become multi-modal. As discussed in the Calibration section, while this is not arbitragible, it is generally not desirable. Figure 2 shows the result of adding a penalty for each PDF minimum proportional to the height difference from its smallest neighbouring maximum. The PDF is now unimodal, and the maximum calibration error of 0.23 basis points in implied normal volatility is still well within bid/ask spreads.

Figure 3 shows a fit to listed SP500 options from November 2023 for the one year expiry. As there are many closely spaced strikes we do not place knots at intervals ΔK_i but rather use a uniform absolute spacing of $\Delta s_i = 0.05$ over the range of quoted strikes. As in the previous example we apply a penalty for PDF minima to constrain the shape of the distribution. The PDF is essentially unimodal with some small interpolation artefact undulations, which are of no practical importance. The maximum error is 0.07% in lognormal volatility, well within the bid/ask spreads.

9 Conclusion

We have introduced the Normal Mapping approach to arbitrage free option price interpolation. The method is flexible, completely analytic, and produces smooth (C^2) option prices. We have shown how to constrain the shape of the resulting PDF during calibration and illustrated how to visualize the mappings in terms of a DLV. We have also discussed arbitrage free interpolation of option prices in the expiry dimension.

¹⁴ x_F^T is still the root of a cubic polynomial.

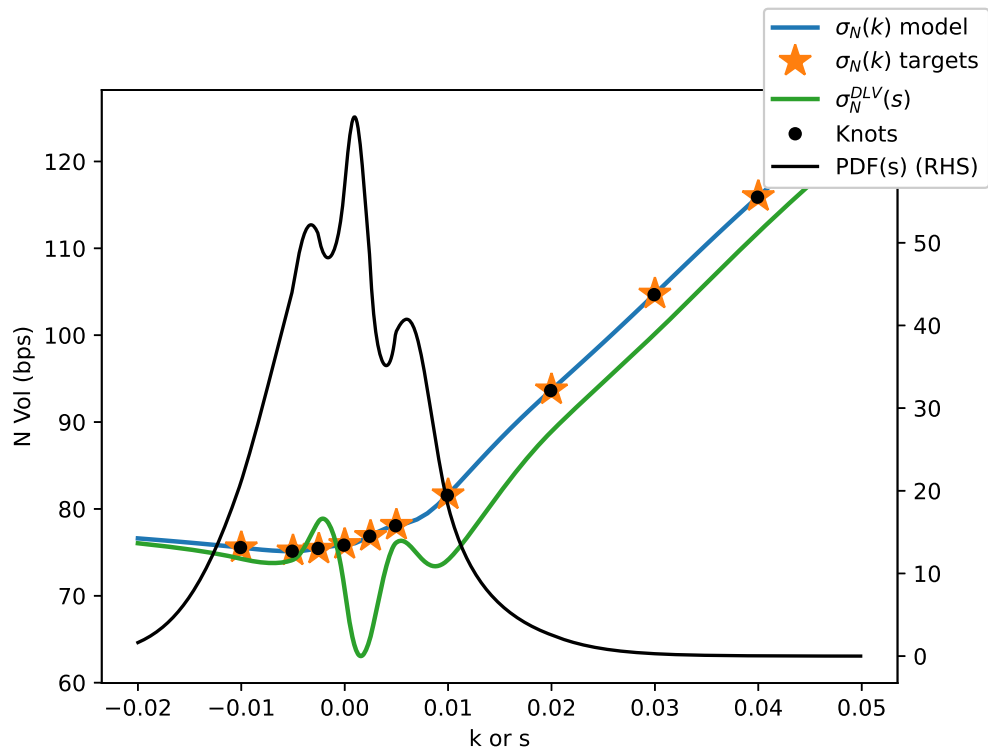


Figure 1: Model implied volatilities fitting to some 1-into-10 USD Swaption data. Targets are repriced to better than 0.1 bps but the PDF is not unimodal.

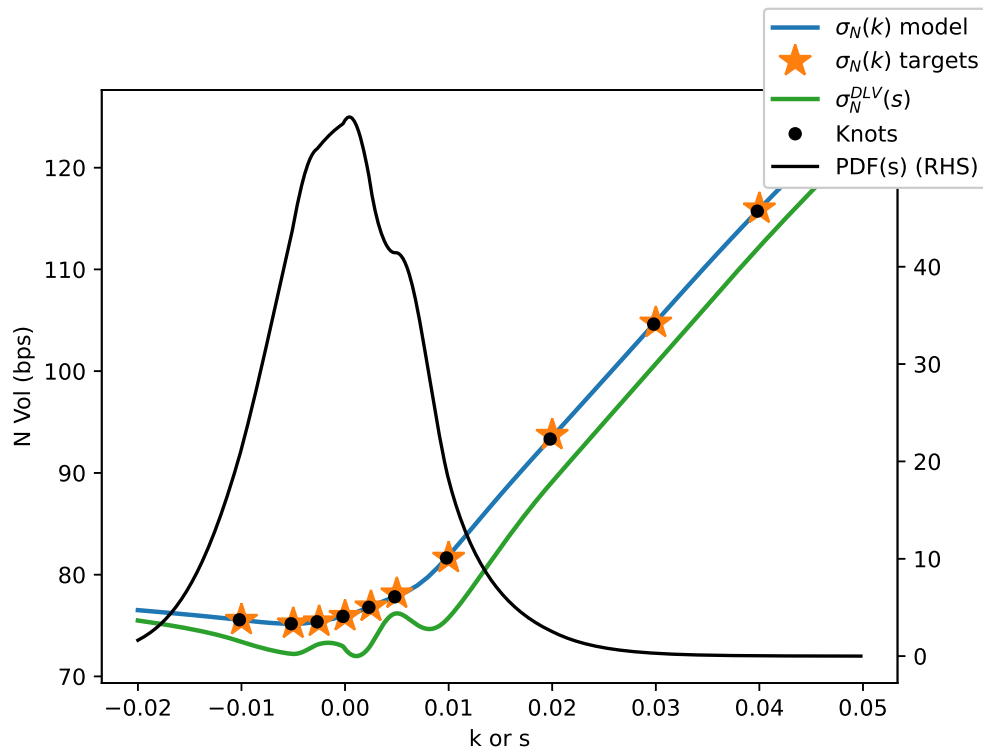


Figure 2: As Figure 1 but with a penalty for PDF minima. The PDF is now unimodal, and the maximum calibration error is less than 0.25 basis points in implied normal volatility.

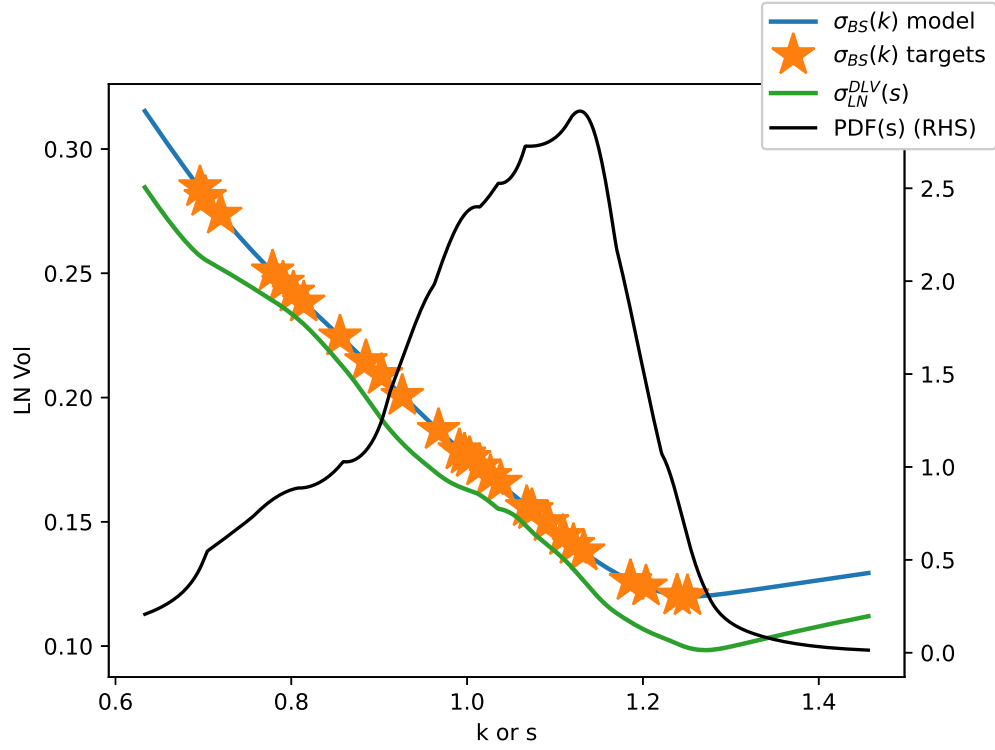


Figure 3: Fit to SP500 option quotes with a penalty for PDF minima. Knots are spaced uniformly with $\Delta s = 0.05$. The maximum calibration error is 7 basis points in implied lognormal volatility.

After completion of this work, we became aware of [11], which obtains the same option price interpolation from a different starting point. While the two studies arrive at broadly consistent conclusions, there are some technical differences. In contrast to our results, the authors of [11] find that calibrating the g_i is more stable than calibrating the x_i (and ultimately settle on using a monotone quadratic B-spline interpolation and optimising the coefficients). As explained in the calibration section, we find that fixing the g_i (or h_i), and therefore the spacing (ratio) of knots in underlying space, produces superior results. This difference may simply be due to differences in the numerical setup, for example the implementation of the forward calibration, our anchoring of the mapping at $g(0) = 0$ ($h(0) = 1$), and the fact that we analytically calibrate the ATM volatility level (exactly in the rates case, approximately for the asset case). Other differences in setup include that we don't exponentiate the mapping in the asset case, and our regularization strategy that directly penalizes local minima in the probability density function.

Aside from these differences, we extend the work of [11] by supporting more general wing extrapolation (and deriving the corresponding implied volatility asymptotics), discussing risk calculation, providing formulas for analytic at-the-money calibration, discussing visualization of the mappings via DLVs, obtaining the short-time expansion of the implied volatility for a general mapping function, and addressing the issue of arbitrage-free interpolation in time to expiry.

Acknowledgements

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Appendix A Monotone Interpolation

The NM approach is defined by specifying increasing sequences $\{x_i\}$ and $\{g_i\}$. To complete the model, we require an interpolation scheme with the following properties: (i) $g(x)$ is guaranteed to be monotone; (ii) $g(x)$ is (at least) C^1 ; (iii) the indefinite integral of $g(x)$ against the normal distribution is known in closed form.¹⁵ Property (iii) rules out more complex schemes like rational interpolators. Property (ii) rules out a simple linear interpolation.

The simplest scheme to fulfill all three properties is *monotone cubic interpolation*. There are a number of variants of this method, the simplest of which share the additional advantages that they are local (i.e. changing g_i impacts the spline only in a neighborhood of x_i), one-pass (i.e. do not require iteration or numerical methods to compute the spline), and depend continuously on the data. A sensible choice for our purpose, and the method used in our numerical examples, is described in [15]. This scheme sets the gradients $\{g'_i\}$ as weighted harmonic means of the neighboring secants, and is implemented in python library SciPy as PchipInterpolator. For an alternative method with similar properties, see [16].

For a somewhat different approach, see the quadratic scheme of [17, 18]. This scheme inserts a knot where required to keep the interpolation monotone, and can also be configured to be shape preserving, i.e. convex where the data is convex and vice versa. Another

¹⁵This is required for closed-form option prices, see (9).

quadratic interpolation, using B-splines with a simple box constraint to enforce monotonicity, is suggested in [11].¹⁶

Appendix B Implied Volatility Asymptotics: Rate Case

Here we obtain the implied volatility asymptotics corresponding to (8). We first note that for $g(z)$ to be increasing, we require $m_l < 0$ and $m_r > 0$. This is generally not a restriction since volatility is stochastic and SV models produce implied volatility smiles (or smirks) that are generally increasing in the wings.

For the right wing, as $z_K \rightarrow \infty$, we have

$$C(K, T) \sim \int_{z_K}^{\infty} m_r T z^2 n(z) dz \quad (28a)$$

$$\sim m_r T n(z_K) (z_K + O(1/z_K)) , \quad (28b)$$

where z_K solves the quadratic $m_r T z^2 + g'(x_n) \sqrt{T} z = k$ as $k \rightarrow \infty$, so $z_K \sim \sqrt{k}$ for $m_r > 0$ and $z_K \sim k$ if $m_r = 0$. On the other hand, the leading term of the Bachelier formula as $K \rightarrow \infty$ is

$$\ln C_N(K, T) \sim -\frac{K^2}{2\sigma_N^2(K, T)T} . \quad (29)$$

We therefore conclude

$$\sigma_N(K, T) \sim \sqrt{m_r K} \quad m_r > 0 \quad (30a)$$

$$\sim \text{constant} \quad m_r = 0 , \quad (30b)$$

i.e. the normal implied volatility grows like the square root of strike (and is proportional to the square root of m_r) if $m_r > 0$, and is asymptotically constant for $m_r = 0$.

Repeating this analysis for the left wing yields

$$\sigma_N(K, T) \sim \sqrt{m_l |K|} \quad m_l > 0 \quad (31a)$$

$$\sim \text{constant} \quad m_l = 0 . \quad (31b)$$

The asymptotes $\sigma_N(|K|) \sim \sqrt{|K|}$ are based on wings of the form (8), where we have set the coefficient of the z^3 term to zero. Including this term leads to faster growth $\sigma_N(|K|) \sim |K|^{\frac{2}{3}}$. If even faster growth is required in the right wing, one can switch to using (13) for the right-wing mapping. As we will see in Appendix E, this leads to growth of the form $\sigma_N(K) \sim K/\sqrt{\ln K}$, the fastest growth allowed by no-arbitrage considerations [19].

Appendix C Quadratic DLV

Here we consider the case of a quadratic Discrete-time Local Volatility, that is we define $g(x) = x\sigma_N^{DLV}(x)$, where

$$\sigma_N^{DLV}(x) = a(x - x_m)^2 + \sigma_m \quad (32)$$

¹⁶Our pricing formulas remain valid for a quadratic scheme, with $a = 0$ in (10). We also note that for quadratic schemes the asset case mapping can be of the form $e^{g(x)}$, since the integral of a quadratic exponential against the normal distribution is known in closed form.

is a quadratic with minimum σ_m at x_m . We want to compute the corresponding implied normal volatilities.

From (9) the exact put price $P(K, T)$ of the model is given by:

$$P(K, T) = kN\left(\frac{x_K}{\sqrt{T}}\right) + (aT(x_K^2 + 2) - 2aTx_m x_K + \sigma_m + aTx_m^2)n\left(\frac{x_K}{\sqrt{T}}\right). \quad (33)$$

where $x_K = g^{-1}(k - \eta)$ and we have used $\eta = -E[g(x)] = 2aTx_m$. On the other hand we have

$$P_N(K, T) = kN\left(\frac{k}{\sigma_N(K)\sqrt{T}}\right) + \sigma_N(K)\sqrt{T}n\left(\frac{k}{\sigma_N(K)\sqrt{T}}\right). \quad (34)$$

where $\sigma_N(k)$ is the normal implied volatility. Let $s_m = x_m(\sigma_m + 2aT)$ be the spot level corresponding to x_m , i.e. $s_m = \eta + g(x_m)$. We now expand (33) and (34) around $K_m = F + s_m$ and equate coefficients. The result is

$$\sigma_N(K_m) = \sigma_m + 2aT \quad (35a)$$

$$\sigma'_N(K_m) = 0 \quad (35b)$$

$$\sigma''_N(K_m) = 2a/(\sigma_m^2 + 2a\sigma_m T). \quad (35c)$$

This shows that the smile of $\sigma_N^{DLV}(s)$ gets translated into a smile of the implied volatility $\sigma_N(K)$, with minimum at $K_m = F + s_m$ and minimum normal volatility equal to $\sigma_m + 2aT$. Figure 4 shows the two smiles $\sigma_N^{DLV}(s)$ and $\sigma_N(K)$ on the same scale, demonstrating that to a good approximation the implied volatility smile is just a vertically displaced version of the DLV.

Appendix D Short-time Expansion

Here we obtain the short-time expansion of the implied volatility under the NM approach. Starting from the equivalent of (9) for the call option

$$C(K, T) = \int_{x_K}^{\infty} \frac{dx}{\sqrt{2\pi T}} e^{-\frac{x^2}{2T}} (\eta + g(x) - k) \quad (36)$$

where

$$\eta = - \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi T}} e^{-\frac{x^2}{2T}} g(x) \quad (37)$$

These are Laplace type integrals with asymptotic expansions¹⁷

$$\eta = g(0) + \frac{T}{2}g''(0) + \dots \quad (38)$$

and

$$C(K, T) = \sqrt{\frac{T}{2\pi}} e^{-\frac{x_K^2}{2T}} \left[\frac{\eta + g(x_K) - k}{x_K} + T \left(\frac{g'(x_K)}{x_K^2} - \frac{\eta + g(x_K) - k}{x_K^3} \right) + \dots \right] \quad (39)$$

¹⁷For simplicity we assume the call is out of the money, i.e. $x_K > 0$.

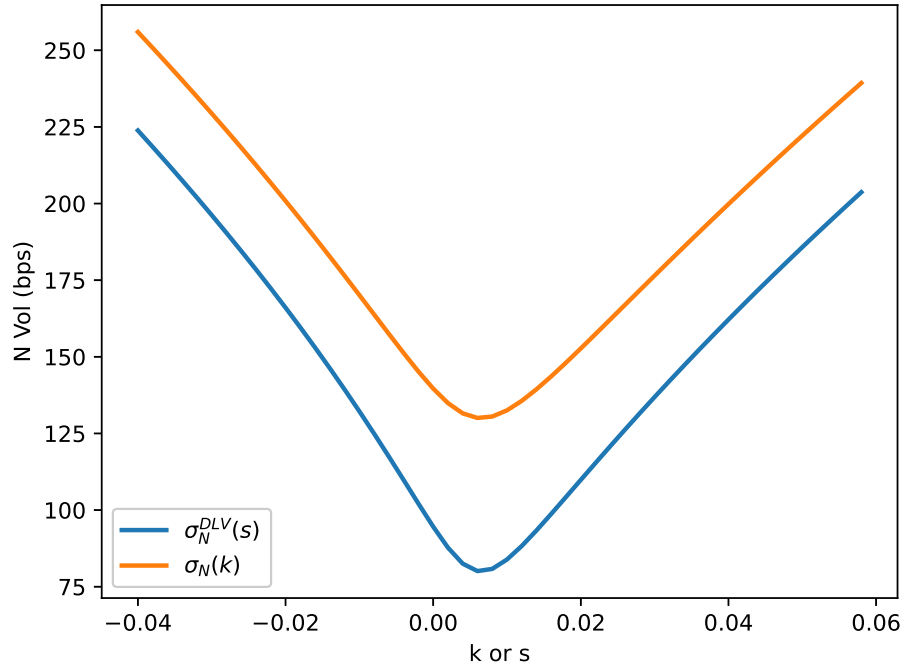


Figure 4: The Discrete-time Local Volatility and corresponding implied volatility for a quadratic DLV in z-space with $a = 0.0025$, $x_m = 0.5$, and $\sigma_m = 80$ bps.

On the other hand, the short time expansion for an OTM Bachelier call with implied volatility $\sigma_N(K)$ is given by

$$C(K, T) = \frac{\sigma_N^3(K) T^{\frac{3}{2}}}{\sqrt{2\pi}(F-K)^2} e^{-\frac{(F-K)^2}{2\sigma_N^2(K)T}} [1 + \dots] . \quad (40)$$

Using $g(0) = 0$, $g(x_K) = k - \eta$, $g(x) \equiv x\sigma_N^{DLV}(x)$, and comparing the coefficients in the leading (exponential) time dependence, we conclude that the zeroth order implied volatility is

$$\sigma_N(K) = \sigma_N^{DLV}(x_K) \quad (41)$$

where x_K is given by $x_K\sigma_N^{DLV}(x_K) = k$. This shows that to leading order in the short expiry limit, the implied volatility at strike $K = F + k$ is just the Discrete-time Local Volatility evaluated at x_K . We use this relation in the calibration section to set up approximate initial conditions for the calibration. Carrying out the expansion to first order we find

$$\sigma(K, T) = \sigma_D(x_k) \left[1 + \frac{T}{x_k^2} \ln \left(1 + \frac{x_k \sigma'_D(x_k)}{\sigma_D(x_k)} \right) - \frac{T \sigma'_D(0)}{x_k (\sigma_D(x_k) + x_k \sigma'_D(x_k))} \right] + O(T^2) \quad (42)$$

We note that this result agrees with the results of Appendix C.

Appendix E Implied Volatility Asymptotics: Asset Case

Here we derive the implied volatility asymptotics corresponding to the asset case extrapolation (13). We follow [20] and [21].

Looking first at the right wing, we define the slope β_r of the right wing implied variance by

$$\beta_r = \lim_{\kappa \rightarrow \infty} \frac{\sigma_{BS}^2(\kappa, T) T}{\kappa} \quad (43)$$

where $\kappa = \ln(K/F)$ and σ_{BS} is the lognormal implied volatility. To determine β_r for our model, we calculate

$$p_r^* = \sup\{p_r : E[s^{1+p_r}] = \int_{-\infty}^{\infty} s(z)^{1+p_r} n(z) dz < \infty\} . \quad (44)$$

On the right wing $s = \eta \exp(m_r(z - z_n)^2 + \dots)$, with $0 \leq m_r < 1/2$,¹⁸ hence the expectation in (44) exists if $p_r < p_r^*$ where

$$p_r^* = \frac{1}{2m_r} - 1. \quad (45)$$

Then $\beta_r = f(p_r^*)$, where $f(x) = 2 - 4[\sqrt{x^2 + x} - x] \in [0, 2]$. For $0 < m_r < 1/2$, p_r^* is finite and the right wing of the implied variance grows linearly with κ . If $m_r = 0$, all moments exist and the extrapolation is flat in implied lognormal volatility. If sub-lognormal implied volatility asymptotics is required, one can use the right-wing behaviour of (8) instead of (13).

¹⁸ $m_r > 0$ so $g(z)$ is increasing and $m_r < 1/2$ so the first moment exists.

For the left wing we similarly define

$$p_l^* = \sup\{p_l : E[s^{-p_l}] = \int_{-\infty}^{\infty} s(z)^{-p_l} n(z) dz < \infty\} . \quad (46)$$

On the left wing $s = \eta \exp(m_l(z - z_0)^2 + \dots)$, with $m_l \leq 0$ required for increasing $g(z)$. We hence have

$$p_l^* = \frac{1}{2|m_l|} . \quad (47)$$

The slope of the left wing implied variance is then given by

$$\beta_l = \lim_{\kappa \rightarrow -\infty} \frac{\sigma_{BS}^2(\kappa, T)T}{|\kappa|} = f(p_l^*) . \quad (48)$$

We see that the implied variance increases linearly with κ for $m_l < 0$, while for $m_l = 0$ all inverse moments exist and extrapolation is flat in lognormal implied volatility.

We note that when β_l or β_r can be estimated from sufficiently out-of-the-money traded option prices, the corresponding slope $m_{l,r}$ can be calculated directly by inverting $\beta_{l,r} = f(p_{l,r}^*)$ to yield

$$p_{l,r}^* = \frac{1}{2\beta_{l,r}} + \frac{\beta_{l,r}}{8} - \frac{1}{2} \quad (49)$$

and then using (45) or (47).

In summary, we have shown that the slopes m_l and m_r control the asymptotic slopes of the implied volatility in a straightforward way. In particular, for non-zero slopes the implied variance grows linearly in $\ln K$. This is the behaviour of asymptotic implied volatility in a large class of stochastic volatility models, including the Heston model, and it is generally considered a desirable feature for an implied volatility extrapolation scheme.

Appendix F Extrema of the PDF

Here we calculate the position and nature of the extrema of the PDF $\phi(s)$. We concentrate on the case without lower bound, but the formulas for the asset case are the same with the replacement $g(z) \rightarrow h(z)$. Starting from (7) we have

$$\phi'(s) = -\phi(s) \left(z + \frac{g''}{g'} \right) \frac{dz}{ds} . \quad (50)$$

Since $\phi(s)$, g' , and dz/ds are larger than zero, the extrema are at

$$zg' + g'' = 0 . \quad (51)$$

This is a cubic equation whose roots are known analytically, so it is simple to check if there are extrema on each interval (z_i, z_{i+1}) . Computing $\phi''(s)$ and using (51) we find that at the extrema

$$\phi''(s)|_{\text{ext}} = \phi(s) \left(-1 + \frac{g'''}{g''} + z^2 \right) \frac{dz}{ds} . \quad (52)$$

Since $\phi(s)$ and dz/ds are positive, the nature of the extremum is determined by the sign of the expression in parentheses, which is easily checked.

This determines the location and type of extrema on the intervals. To complete the analysis, we must also check the knot points themselves, which can be extrema since g'' is potentially discontinuous there. To do this, we simply check if (50) changes sign at the knot point.

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